Boolean Numerical P System-Based Model for Cloud Workload Prediction

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Weekly Review Report

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Week	Start Date - End Date	Work carried out during the week (with Your signature and Date)	Internal guide's comments with signature and Date
1	5/01/2026 - 11/01/2026	Problem Statement Identification B S Reddy 12/01/26	
2	12/01/2026 - 18/01/2026	Conducted Literature Survey B S Reddy 19/01/26	
3	19/01/2026 - 25/01/2026	Defined BNPS formal tupule and outlined 3 phase execution semantics B S Reddy 26/01/26	
4	26/01/2026 - 01/02/2026	Implemented serial BNPS simulator with .pep input parsing B S Reddy 2/2/26	
5	02/02/2026 - 08/02/2026	Developed CUDA kernel code for multi membrane B S Reddy 9/2/26	
6	9/02/2026 - 15/02/2026	Encoded single membrane linear regression pipeline B S Reddy 16/2/26	[illegible] 3/3/26
7	16/02/2026 - 22/02/2026	Built logistic regression witn sigmoid approximation on synthetic data B S Reddy 23/2/26	
8	23/02/2026 - 01/03/2026	Tested 21 membrane Iris logistic regression B S Reddy 2/3/26	
9	02/03/2026 - 08/03/2026	Review Week: report consolidation and mid-semester presentation preparation B S Reddy 3/3/26	
	Mid Semester Review Feedback		

Introduction

- Membrane Computing (MC) models cell-like parallel computation
- P Systems: computational model with parallel membrane regions
- Numerical P Systems (NPS) use real-valued arithmetic variables
- BNPS: NPS extended with Boolean conditional operators (AND, OR, NOT, XOR)
- Goal: encode ML algorithms natively within BNPS framework

Problem Definition

- No BNPS simulator supporting Boolean conditions exists currently
- No GPU-parallel multi-membrane BNPS simulator available
- Gradient-descent ML encoding in BNPS not yet attempted
- Cloud workload prediction unexplored within membrane computing
- Need: BNPS-based cloud resource utilisation prediction model

Literature Survey

S.No.	Authors (Citation)	Year	Dataset / Approach Used
1	Păun [1]	2000	P Systems: theoretical membrane computing foundation
2	Păun & Păun [7]	2006	Numerical P Systems (NPS): variables, programs, repartition
3	Pavel et al. [10]	2010	Enzymatic NPS (ENPS): enzyme-driven production rules
4	Liu et al. [15]	2019	BNPS: Boolean condition guards on NPS programs
5	Raghavan et al. [22]	2020	GPUPeP: CUDA-parallel NPS simulator; GPU vs CPU benchmark
6	Raghavan et al. [23]	2019	Multi-ENPS: multi-membrane simulator with file inter-conversion
7	Nehra & Kesswani [24]	2024	Cloud workload prediction; SLA violation reduction model
8	Ouhame et al. [25]	2021	CNN-LSTM for cloud resource utilisation forecasting
9	Zhao et al. [26,27]	2025	TFEGRU/MSCNet: cloud workload prediction benchmarks

Work Done

- Designed BNPS simulator: serial (Python) and GPU-parallel (CUDA)
- Encoded linear regression in BNPS using 7-stage pipeline architecture
- Piecewise-linear sigmoid approximation for logistic regression in BNPS
- Multi-membrane logistic regression validated on Iris dataset (21 membranes)
- Achieved 1.94x GPU speedup for 21-membrane configuration

Work Done

BNPS Simulator

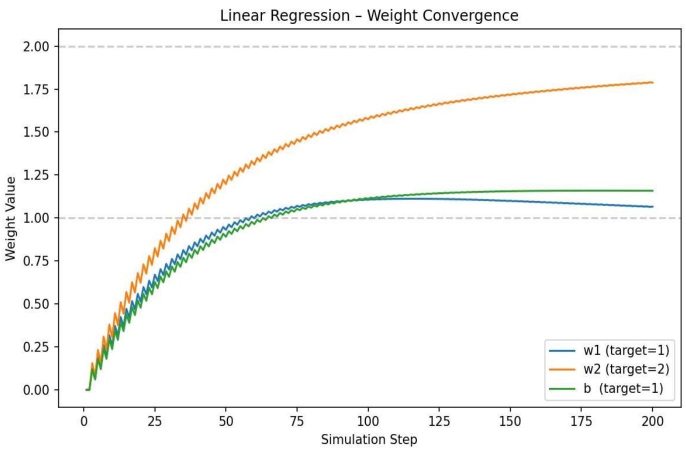
- $\Pi = (m, \Pi, \mu, (Var_1, Var_1(0)), \ldots, (Var_m, Var_m(0)), Pr)$
- Var_i = variables of membrane i; Pr = programs controlling dynamics
- Program form: $F(\text{variables}) \mid C(\text{variables}) \rightarrow c_1|v_1 + c_2|v_2 + \cdots + c_n|v_n$
- Three-phase: evaluate C(vars), compute F(vars), redistribute outputs

ML Algorithms in BNPS

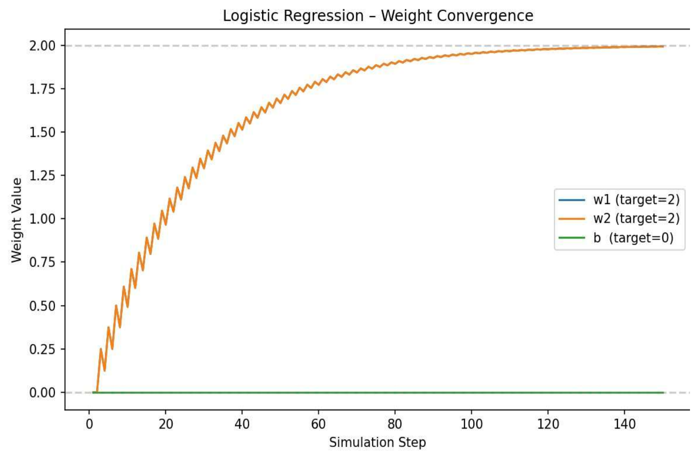
- Linear regression: $\hat{y} = w_1x_1 + w_2x_2 + b$ trained via Batch GD + MSE
- Logistic regression: sigmoid $\sigma(z) \approx 0.5 + 0.25z$ (piecewise-linear, BNPS-compatible)
- Multi-membrane: 1 controller + 20 sample membranes; weights ↓, gradients ↑
- Boolean convergence guard halts updates when gradient reaches zero

Analysis/ Results

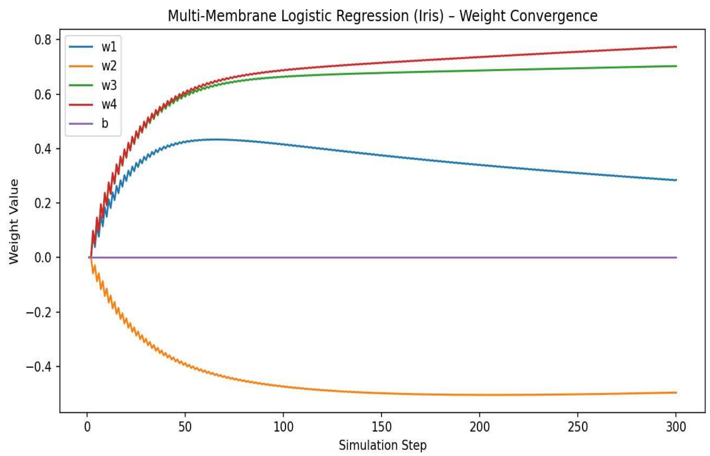
Linear Regression



Logistic Regression

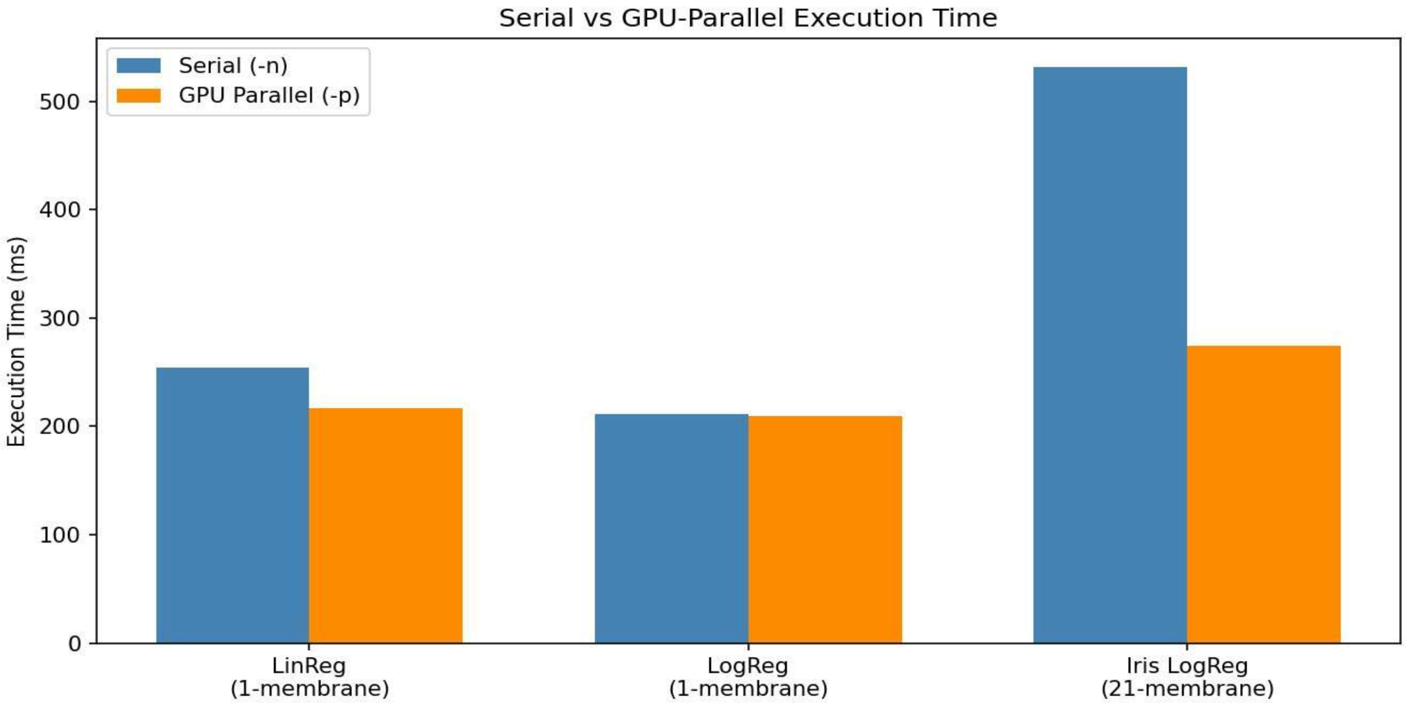


Iris Multi-Membrane (21)



- LinReg: w1~1.07, w2~1.79, b~1.16 converging to true values (1, 2, 1)
- LogReg: w1=w2 reach 2.0; bias b stays at 0 due to symmetric dataset
- Iris: petal features (w3, w4) dominate; w2 negative reflects Setosa sepal width

Analysis/ Results



Key Results

- LinReg (1-mem): 1.17x speedup
- LogReg (1-mem): 1.01x speedup
- **Iris (21-mem): 1.94x speedup**
- GPU gains scale with more membranes
- Serial and GPU outputs identical

Conclusion

- First BNPS simulator supporting Boolean conditions and GPU-parallel execution
- Linear and logistic regression natively encoded; weights converge correctly
- Multi-membrane architecture validated on real-world Iris dataset successfully
- GPU parallelism scales effectively with increasing membrane count (1.94x)
- Feasibility of gradient-descent ML within membrane computing demonstrated

Future Work

- Encode a BNPS-based model specifically for cloud workload prediction
- Extend the framework to process real resource utilisation data from cloud service providers
- Forecast anticipated CPU, memory, and network demands using BNPS
- Leverage the inherent parallelism of the multi-membrane architecture for efficient computation
- Achieve prediction accuracy comparable to traditional deep learning approaches

References

1. G. Păun, “Computing with Membranes,” Journal of Computer and System Sciences. 2000.
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7. P. Nehra and N. Kesswani, “A Workload Prediction Model for Reducing Service Level Agreement Violations in Cloud Data Centers,” 2024.
8. S. Ouhame et al., “An Efficient Forecasting Approach for Resource Utilization in Cloud Data Center Using CNN-LSTM Model,” Neural Computing and Applications,2021

Thank You

Any Questions?